

A rapid indicator of cultural ecosystem service usage at a fine spatial scale: Content analysis of social media photographs



Daniel R. Richards*, Daniel A. Friess

Department of Geography, National University of Singapore, 1 Arts Link, Singapore 117570

ARTICLE INFO

Article history:

Received 17 October 2014

Received in revised form 19 January 2015

Accepted 22 January 2015

Keywords:

Ecosystem service assessment

Recreation

Mangrove

Content analysis

Flickr

ABSTRACT

Cultural ecosystem services (CES) are an important benefit that habitats provide, particularly in the fragments of natural ecosystems that remain in urban areas. To manage CES we need to understand what people use habitats for, and where different activities take place. It is challenging to assess CES provision, as surveys and interviews are time consuming and can be expensive. Social media data, particularly geo-tagged photographs, are spatially explicit and contain visual information that can be used to infer cultural use. Indicators of CES derived from social media make use of existing data so may contribute useful information for rapid, cost-effective assessments of CES. In this study we develop an indicator of CES usage that is derived from photographs from an image-sharing website, at two different scales. First, we compare four small (<150 ha) urban mangrove sites in Singapore, using photograph content to classify sites according to predominant cultural use. Second, the spatial distribution of different CES was modelled within one site using MaxEnt. A resampling simulation was conducted to identify the sensitivity of the photograph classification to the number of photographs classified. Photographs of social recreation, organisms and landscapes occurred most commonly. The proportional occurrence of photograph types differed between sites depending on their characteristics. Within one site, the probability of occurrence of social recreation photographs was highest around built focal points, while photographs of organisms were more likely in the mangrove and terrestrial habitats. Classifying more than 50–70 photographs (which would take approximately 30 minutes) gave only small increases in categorisation accuracy. This tool for CES assessment rapidly provided information that would be useful for managing Singapore's mangroves. The approach could be widely applied to assess CES provision across a range of habitats and settings, helping CES to become more commonly considered in ecosystem service evaluations.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

The concept of ecosystem services – the benefits that people gain from nature (MA, 2005) – has been proposed as a framework for managing and conserving habitats. An ecosystem can provide a range of provisioning, regulating, supporting and cultural services, and the subset of services that are provided is affected by the way that people manage the environment. There are commonly trade-offs between the provision of different ecosystem services, so decisions must be made about how to balance their provision (Bennett et al., 2009), an understanding of which will help managers make more informed decisions (Fish, 2011). For ecosystem service frameworks to become widely used in environmental decision making, it must be straightforward and cost-effective to analyse service provision, especially at the local scales (e.g. within

a nature reserve) at which many decisions are made (Peh et al., 2013). Multiple methodological and modelling frameworks have now been developed to assist with ecosystem service assessment [e.g. ARIES (Villa et al., 2009), InVEST (Guerry et al., 2012), TESSA (Peh et al., 2013), and the benefit transfer approach applied by Burkhard et al. (2009)]. Such frameworks typically focus on provisioning and regulating services, with fewer options available for assessing cultural ecosystem services (CES) (Crossman et al., 2013).

CES are a broad subset of services that are nonmaterial, such as the existence value of a species, or the opportunities for wildlife recreation, social recreation in public spaces, or education that habitats can provide (Hernández-Morcillo et al., 2013). When assessing CES usage there is a trade-off between the level of detail (both spatial and in terms of differentiating between CES) that an assessment can provide, and the amount of time required to make the assessment. Traditional sociological approaches such as surveys (Pleasant et al., 2014), interviews (Plieninger et al., 2013) and focus groups (Norton et al., 2012) can provide detailed information on relative CES usage, but are time consuming to carry out and

* Corresponding author. Tel.: +65 65163851.

E-mail address: d.r.richards@nus.edu.sg (D.R. Richards).

Table 1
Summary information describing the four study mangroves in Singapore.

Site	Protection status	Approximate mangrove area (ha)	Accessibility	Key amenities
Berlayer Creek	Nature Reserve ^a	3	Public access, free, good transport link	Playground and fitness, historical sites, mangrove and ocean boardwalks
Chek Jawa	Nature area ^b , previously threatened by reclamation	2	Public access, free, longer journey from mainland followed by hike, bicycle, or bus	Mangrove and ocean boardwalks, viewing tower, adjacent coastal forest, seagrass
Sungei Buloh Wetland Reserve (SBWR)	Nature Reserve ^a – designated for birdlife	133	Public access, free, longer journey from main urban areas	Nature trail and wildlife information, viewing points, bird hides, mangrove boardwalk
Mandai	State land – no legal protection	3	No public access	None

^a Highest legal conservation protection status in Singapore.

^b Managed by the governmental National Parks organisation. Unclear legal protection.

rarely provide spatially explicit information (Hernández-Morcillo et al., 2013). More recent indices derived from geographic data have been proposed for mapping CES rapidly and with high spatial accuracy over wide geographic areas. For example, the volume and spatial distribution of geo-tagged photographs that have been uploaded to social media may indicate recreational value (Wood et al., 2013) and, by extension, aesthetic quality (Casalegno et al., 2013; Nahuelhual et al., 2013). This approach has now been implemented in InVEST, a popular modelling and mapping framework for ecosystem services assessment (InVEST, 2014).

Existing social media indices of visitor pressure as quantified by photograph density can be used for broad-scale analyses of CES provision, as they indicate popular areas for visitors. However, photograph density does not help us to assess the cultural role of a site, or the reasons why it may be appreciated by society. To add this important detail for site management, we must understand the activities that people are engaged in when they take photographs, and which features of the environment attract their attention. Content analysis of social media photographs could allow indicators of individual CES to be identified, and their spatial patterns to be mapped. Content coding (classification) of photographs taken in the field by survey subjects has previously been used to identify factors that impact recreational enjoyment (e.g. Dorwart et al., 2010). However, such a classification has not been applied to online datasets of existing geo-tagged photographs.

The aim of this study is to present a methodological framework for assessing different CES uses through the content analysis of photographs, using the freely accessible photo-sharing website Flickr. We analyse photograph content at two spatial scales (between sites and within-site) to compare an indicator of CES provision in the context of coastal mangrove forest habitat patches that remain in Singapore. The specific objectives of this study were to: (1) compare CES provision between four mangrove sites in Singapore, by downloading and classifying images hosted on Flickr; (2) analyse spatial variation in cultural usage within one mangrove site, by modelling the distribution of different types of photographs in relation to habitat characteristics and; (3) assess the time required to implement this method, by analysing the sensitivity of the photograph classification to the number of photographs classified.

2. Materials and methods

2.1. Case study system: mangroves in Singapore

Singapore is a suitable location to analyse social media photographs due to a high population density of ~7600 people per

km² (Singapore Department of Statistics 2013), and high rate of mobile phone usage. In 2012 there were around 1.5 phones per person (Singapore Department of Statistics 2013; IDA, 2014), many of which are smartphones that are capable of capturing and uploading geo-tagged photographs. Singapore is also a useful location to understand the importance of CES in mangrove conservation and management, as the city-state has undergone rapid urbanisation in recent decades, with a concomitant areal decline in key habitats such as mangrove forests (Yee et al., 2010). Current patches of mangrove habitat in Singapore are small and fragmented, and variable in terms of their public accessibility and their amenities (Table 1).

We analysed CES provision at four mangroves in Singapore; two nature reserves, one natural area, and one undeveloped area of state land (Table 1). Three of the sites are located on Singapore's mainland, with a further site on the offshore island of Pulau Ubin (Fig. 1b). The area of mangrove at each site is relatively small, and because mangrove habitat is not found independently, adjacent habitats such as mudflat, seagrass, coastal forest and open ocean were also assessed. All but one of the mangrove sites are publicly accessible (Table 1), but there is variation in the travel time from the main urban areas (Table 1). One site, Mandai, is not accessible to the general public but is utilised extensively for academic research purposes (Friess et al., 2012). There is variation in the amenities that are available at, or adjacent to, the sites (Table 1). Berlayer Creek is set within a recreational park environment, while Sungei Buloh Wetland Reserve (SBWR) and Chek Jawa have more educational content (Table 1).

2.2. Assessment workflow

This study presents a methodological framework for differentiating between CES uses at a fine spatial scale, using social media data. The resulting index of CES use may be used within a larger CES assessment to assist habitat management. Under this framework, photographs must first be accessed from Flickr using a geographic search. Second, the content of the photographs must be classified using an objective coding approach that distinguishes between different types of CES usage by the public. The proportions of different types of photographs can then be compared between sites. Finally, the detailed geospatial data attached to each image can be used within a distribution modelling tool such as MaxEnt to investigate the relationships between environmental factors and cultural usage, and map indices of spatial variation in different cultural uses across a single mangrove site.

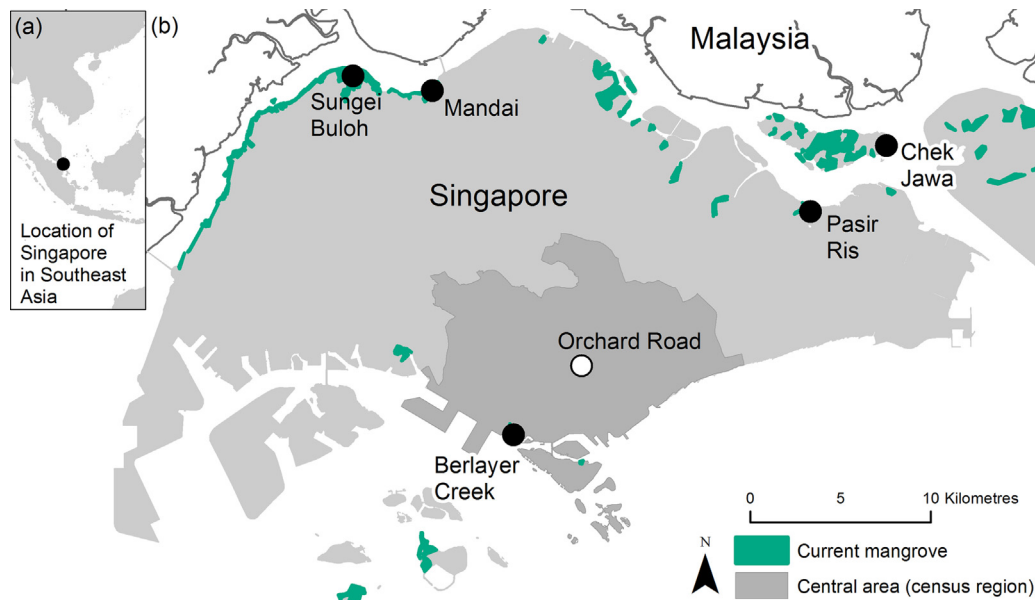


Fig. 1. (a) Location of Singapore in Southeast Asia. (b) Map of indicative mangrove area and study sites in Singapore. Mangrove area was estimated from Yee et al. (2010).

2.3. Analysis of Flickr images

2.3.1. Collection of Flickr images

Flickr is a major image-sharing website, with >70 million users and 200 million photographs uploaded worldwide (Wood et al., 2013). It is not clear how many Flickr users are based in, or have visited Singapore, but the website is likely to have a reasonable market share in the country when compared to its most similar competitors, such as Photobucket (www.photobucket.com), Panoramio (www.panoramio.com), and Jiebang (www.jiebang.com) (Fig. 2). The website Instagram appears to have a much higher degree of usage in Singapore (Fig. 2), but cannot be used as the application programming interface (API) forbids automated processing of data without permission from users (Instagram, 2014).

Flickr was accessed through the publicly available API, and a rectangular area around each of the mangrove sites was searched for photographs. R code and instructions for sampling Flickr photographs can be found in Appendix S1 in the online supporting

information. The Flickr API returns the most recent 250 photographs (if this many are available) that were geo-tagged within a given geographic search area (Flickr, 2014). The positional accuracy of the location data is likely to vary depending on the device that was used to record the location, and the weather conditions that were present. However, geotagging using most mobile phones is accurate to at least 10 m (Zandbergen and Barbeau, 2011).

2.3.2. Interpretation of Flickr images and analysis of classification reliability

The content of the Flickr images was analysed by eight students or professionals in fields related to environmental management. Each person independently classified all photographs returned by the Flickr search. The ratio of female to male classifiers was 5:3 and four nationalities were represented: Singapore, Taiwan (ROC), United Kingdom and China (PRC). The photographs were coded based on objective criteria that related to the primary subject of the photograph, as opposed to broad, subjective ecosystem service categories such as “recreation” or “aesthetic value”. The categories therefore refer to cultural uses of the ecosystem, rather than ecosystem services per se; in Section 4 we consider how these objective photograph categories may be related back to the typology of CES used by the MA (2005). The coding categories were not developed as an exhaustive list of all those that may be encountered in all habitat types, but are specific to the study system. However, several of them, such as “nature appreciation”, “landscape” and “social recreation” are general categories that may be expected in many systems. Each photograph was first screened to establish whether or not it related to an area of mangrove, and was then classified into one of nine categories using the following criteria:

- (1) Nature appreciation – photographs that primarily depicted animals or plants;
- (2) Landscape – photographs that depicted a wider view and did not focus on a specific feature;
- (3) Social recreation – photographs that depicted people in informal situations with sports or no equipment. Photographs that depicted recreation equipment (e.g. bicycles) but no people were also included in this category;
- (4) Fishing recreation – photographs that depicted people engaged in fishing or crabbing;

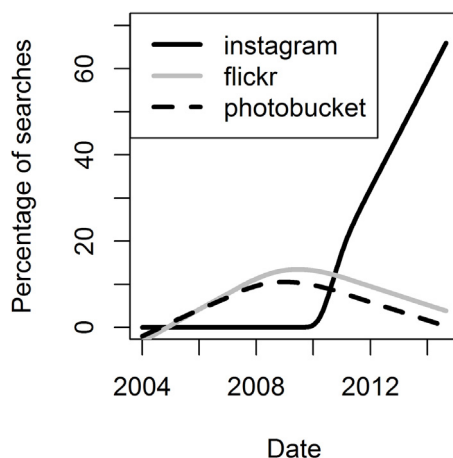


Fig. 2. Smoothed (lowess) relative number of Google searches for the terms “instagram”, “flickr”, and “photobucket” in Singapore, between January 2004 and August 2014. Values are a percentage of the highest recorded search volume for any of the three terms. Comparable searches for “panoramio”, and “街旁” (jiebang) never made up more than 0% of the highest recorded score so are not shown.

- (5) History – photographs that primarily depicted historical objects (e.g. gravestones, building remains);
- (6) Research – photographs that depicted people with research or educational equipment (e.g. clipboards, location markers). Photographs that showed research equipment but no people were also included in this category;
- (7) Infrastructure appreciation – photographs that primarily depicted aspects of the built environment (e.g. boardwalks);
- (8) Other – this category was made up from photographs that primarily depicted non-natural debris items, or soil. The CES usage of these photographs was not clear so they were categorised separately.

The reliability of the classification between people was assessed using Fleiss' kappa as implemented in the *irr* package for R (Gamer et al., 2012). Fleiss' kappa indicates the agreement between the different people conducting the classification, and is a measure of agreement that is commonly used to assess the reliability of content analysis (Hayes and Krippendorff, 2007). The strength of the agreement according to the kappa value was assessed following Landis and Koch (1977). Modal classifications of each photograph (based on the classification by the eight people) were used for the rest of the study.

2.4. Analysis of variability in cultural uses between mangrove sites

The proportion of photographs in each CES category was compared between the four sites using a Chi-squared contingency table. Differences in proportional occurrence of photograph types between sites were considered qualitatively and are discussed in relation to the habitat, infrastructure and human characteristics of the four mangroves, which are briefly outlined in Table 1.

2.5. Analysis of spatial patterns in CES provision within a single mangrove site: Chek Jawa

To assess the spatial segregation of CES provision within a single mangrove, we focused on one of the case study sites which had a comparatively even balance of a high number of photographs from several CES categories (Chek Jawa). We modelled the probability of presence of the three most commonly occurring types of photograph at this site. Each type of photograph was modelled in response to four environmental factors: (1) distance from nearest footpath (including the boardwalk), (2) distance from focal points (rest shelters and a viewing tower), (3) distance from the site entrance, and (4) the dominant habitat type within the neighbouring 30 m (either mangrove, coastal forest, or ocean). Habitat type was classified from a four-band Pleiades satellite image (pixel size = 4 m², taken between April 2012 and August 2013). The accuracy of the classification as assessed using 30 ground reference points was high (93% correctly classified, Cohen's kappa = 0.9). The locations of footpaths, the site entrance and focal points were digitised from the Pleiades satellite imagery.

The probability of the presence of each type of photograph was modelled independently using maximum entropy (MaxEnt), a technique based on machine learning which is commonly used to model species distributions using presence-only data (Elith et al., 2011), but which has also been applied to model the distribution of cultural ecosystem services (Sherrouse and Semmens, 2012). All data were converted to grid (raster) formats at the same resolution as the habitat dataset (pixel size = 4 m²). The quality of MaxEnt models is typically assessed using the area under the curve (AUC) of the receiver operating characteristic (ROC) plot. The AUC score can range between 1, indicating perfect prediction of the presence of photographs, and 0.5, indicating that the model predictions are

no better than random. MaxEnt models with AUC scores greater than 0.75 are generally considered acceptable (Phillips and Dudík, 2008). We trained the MaxEnt models using randomly sampled subsets (80%) of the datasets, and used the remaining 20% of the datapoints to test the predictive accuracy of the models. The AUC for the model run that includes the test dataset assesses the ability of the model to correctly predict new data. To visualise the effects of distance from focal points and habitat type on the occurrence of photographs, variable response curves for these factors were generated from the models for each type of photograph. For response curve prediction, the models were held constant for the non-target variables in the model: at values of 0.5 m from the footpath, 200 m from the entrance, 10 m from the focal points, and within mangrove habitat.

2.6. Sensitivity of site-assessed CES to the number and sequence of photographs classified

Rapid assessments should accurately characterise the CES provision of a site while minimising sampling effort (i.e. classify a minimal number of photographs). It is therefore useful to know how many photographs we need to classify to be reasonably confident that the suite of CES provided by a site has been characterised accurately. In this study, we define an "accurate categorisation" as one in which at least 95% of the photographs are classified in the categories that would be expected given the whole sample.

Different levels of sampling effort were simulated using a bootstrap resampling method. Levels of sampling effort were simulated in increments of one, from the categorisation of one photograph only to the inclusion of all mangrove images in the dataset (52 images) in the case of Berlayer Creek, and to a maximum 100 photographs for the other sites. For each iteration, a subset of the total photographs classified at each site were sampled at random. The proportions of the sample photographs that were classified into each of the nine categories were then compared to the proportional occurrence of each category in the whole sample. The absolute differences in the proportions of photographs classified in each category, between the whole sample and resampled subset, were calculated. The average absolute difference in proportions across the seven photograph categories was taken as an index of similarity. This was repeated 1000 times for each level of sampling effort, and the mean similarity and 95% bootstrap confidence intervals were calculated. The lowest number of photographs that would give a correct classification at least 95% of the time was recorded as the minimum sample size required.

The sequence of the photographs returned by a Flickr search may also affect the classification of a site, as there may be non-random patterns in the types of photograph that result from seasonality or photographer bias. To investigate the sensitivity of the classification to the sequence of the photographs we also plotted the cumulative similarity of classification for the sequence of the first 100 photographs returned by the Flickr API (52 in the case of Berlayer Creek).

3. Results

3.1. Comparison of CES indicators observed in photographs between mangrove sites

Flickr searches returned 760 photographs from within the study sites, of which 683 related directly to mangroves or associated marine habitats (Table 2). The number of suitable photographs varied by site (Table 2). Photographs were taken between 2012 and 2014, with some variation in dates between the sites (Table 2). Individual Flickr users took on average relatively few photographs at

Table 2
Summary of photographs collected by Flickr searches at five mangrove sites in Singapore.

Site	Total number of photos returned by Flickr	Number of mangrove photographs	Date range	Number of photographers	Average number of photographs per visitor
Berlayer Creek	65	52	2012–2013	24	2.17
Chek Jawa	248	216	2011–2014	203	1.06
SBWR	205	227	2013–2014	150	1.51
Mandai	244	188	2014–2014	22	8.55

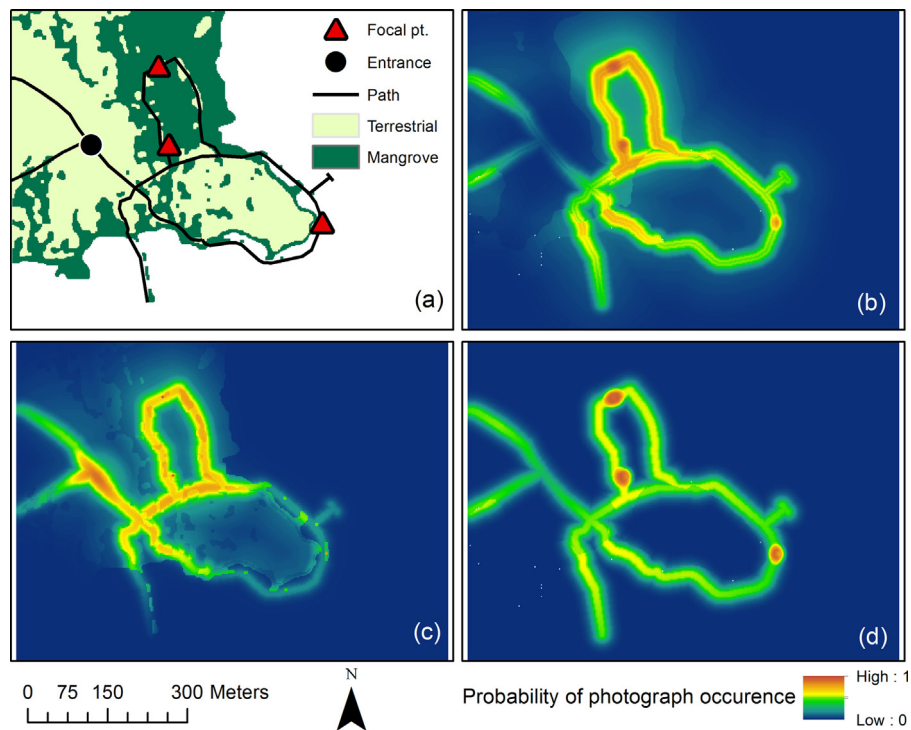


Fig. 3. Modelled areas of high suitability for taking three types of photographs at Chek Jawa. (a) Study site with footpaths, focal points and boardwalks marked, (b) probability of occurrence of landscape photographs, (c) probability of occurrence of nature appreciation photographs, (d) probability of occurrence of social recreation photographs. Probabilities were modelled using MaxEnt.

each site, except in the case of Mandai, where each photographer uploaded on average 8.5 geo-tagged pictures (Table 2). There was substantial agreement in classification between the eight people who classified the photographs (Fleiss's kappa = 0.75).

The proportional occurrence of different types of photographs varied significantly between sites (Chi-squared = 436.8, d.f. = 32, $p < 0.001$). Large proportions of nature appreciation photographs (close-ups of either plants or animals) and considerable proportions of landscape photographs were taken at all sites, and social recreation was also a frequently occurring category at three sites (Berlayer Creek, Chek Jawa, and Mandai) (Table 3). At Berlayer Creek, a Nature Reserve with a modern boardwalk, the dominant

categories were nature appreciation, social recreation, infrastructure and landscape (Table 3). At Chek Jawa, a site that contains multiple marine habitats and has a viewing tower and boardwalk over the sea, the most frequently occurring photographs were of landscapes, social recreation, and nature appreciation (Table 3). At SBWR, a Nature Reserve and wetland education centre, nature appreciation was the most frequently occurring type of photograph, although around a quarter of photographs focused on the landscape (Table 3). At Mandai, a mangrove that is not publicly accessible, the majority of photographs were of research activities (Table 3).

3.2. Spatial segregation of CES types within a single site: Chek Jawa

The probability of occurrence of the three most commonly occurring types of photographs was spatially modelled at Chek Jawa; landscape photographs ($n = 63$), social recreation photographs ($n = 59$) and nature appreciation photographs ($n = 52$). The AUC was high for all models; for the landscape photographs the training AUC was 0.94 and the test AUC was 0.96, for the social recreation photographs the training AUC was 0.96 and the test AUC was 0.94, and for the nature appreciation photographs the training AUC was 0.94 and the test AUC was 0.96. Distance from the

Table 3
Proportions of photographs classified into nine descriptive categories.

Category	Berlayer Creek	Chek Jawa	SBWR	Mandai
Nature appreciation	0.46	0.24	0.64	0.24
Landscape	0.13	0.29	0.22	0.15
Social recreation	0.17	0.27	0.06	0.04
Research	0	0	0	0.35
Infrastructure appreciation	0.23	0.18	0.06	0.01
Fishing	0	0.01	0	0.02
History	0	0	0.01	0
Other	0	0.01	0	0.19

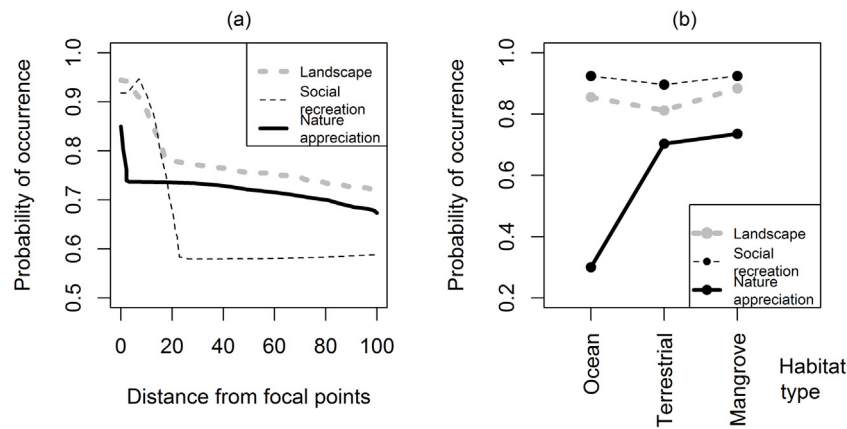


Fig. 4. Variable response curves for two variables (distance from focal points, and majority habitat type within 30 m) for the MaxEnt models describing the distribution of landscape, social recreation, and nature appreciation photographs.

footpath routes was the most important explanatory factor for all types of photograph, indicating that photographs were most likely to be taken on or very close to the footpath (Table 4; Fig. 3). Distance from the focal points had the next greatest impact on probability of occurrence of landscape and social recreation photographs (Table 4), and these photographs showed similar distribution patterns, with peaks in the probability of occurrence in the close vicinity of focal points (Fig. 3; Fig. 4a). Nature appreciation photographs showed a far smaller response to proximity to focal points (Fig. 3; Fig. 4a), but were spatially segregated by the majority habitat type within 30 m (Table 4; Fig. 3); they were far less likely to occur in ocean-dominated areas (Fig. 4b). The probability of occurrence of social recreation and landscape photographs differed little depending on the surrounding habitat type (Table 4; Fig. 4b).

3.3. Sensitivity of site-level CES assessment to the number of photographs classified

In the case of all mangrove sites, the relationship between the number of photographs classified and the accuracy of the site-level characterisation was positive and nonlinear, with the rate of improvement slowing to reach an asymptote (Fig. 5a and d). The minimum numbers of photographs necessary to categorise in order to reach 95% agreement with the full dataset were 41 for SBWR, 46 for Berlayer Creek, 56 for Chek Jawa and 64 for Mandai. The maximum time taken to classify the entire dataset of 682 photographs was approximately 5 h, so it should be feasible for one assessor to classify 70 photographs (enough to reach 95% similarity at any of our study sites) in around 30 min. The sequential classification of photographs resulted in some instability in the similarity of the categorisation to the whole sample classification (Fig. 5a and d). This was particularly true for Berlayer Creek and Mandai, where there were noticeable peaks and troughs in similarity (Fig. 5a and c). However, the similarity to the whole-sample classification

was consistently greater than 0.9 after classifying 20 photographs (Fig. 5).

4. Discussion

Previous CES studies that have used photographs from social media have treated them as homogenous indicators of cultural interest, but content analysis of the photographs allows different cultural uses to be identified. This study compared the CES usage of relatively small patches of habitat (Table 1), and mapped the spatial distribution of 3 CES at Chek Jawa over less than 10 ha. These levels of spatial detail are compatible with the scale of much urban environmental management, which commonly considers single sites (Peh et al., 2013).

4.1. Variation in the CES provided by different mangrove sites

The classification of Flickr photographs revealed segregation in CES usage between the mangroves. The CES usage of sites is likely to differ depending on (1) the physical and ecological features that are present, and (2) social and demographic factors, including the location and surroundings of the site, its accessibility and perception with the public.

SBWR is a Nature Reserve, with abundant and diverse animal and plant life. It is likely that the largest proportion of animal and plant photographs was taken at SBWR because the opportunity for taking these photographs is greater. Three types of photographs were similarly abundant at Chek Jawa, indicating a more diverse range of CES. This was possibly because of the habitat diversity of the site; dense mangrove and coastal forests can be suitable for photographing plants and wildlife, while the boardwalk over the open ocean may provide good views of the surrounding islands.

The public perception and accessibility of a nature site may also impact the characteristics of the people that choose to visit and, indirectly, the types of photographs that are taken. SBWR is internationally recognised as an important site for migratory birds (Table 1; EAAFP, 2012), so may attract a higher proportion of visitors who are interested in wildlife viewing. Chek Jawa is located on the island of Pulau Ubin, which is perceived primarily as a destination for tourism and recreation (Henderson, 2000). Due to its location in a recreational hotspot, some visitors may enjoy Chek Jawa as part of a social visit, while others may visit specifically to observe wildlife. Research photographs were recorded at Mandai because public access is restricted, reducing recreational and nature appreciation visitors.

Table 4
Permutation importance of predictors in the MaxEnt models for the probability of occurrence of landscape, social recreation, and nature appreciation photographs at Chek Jawa (percentage contributions to models).

Factor	Landscape	Social recreation	Nature appreciation
Distance from path/boardwalk	76.5	96.3	61.5
Distance from site entrance	14.7	3.2	7.1
Main habitat type within 30 m	0	2.3	17.2
Distance from focal point	8.8	1.9	14.2

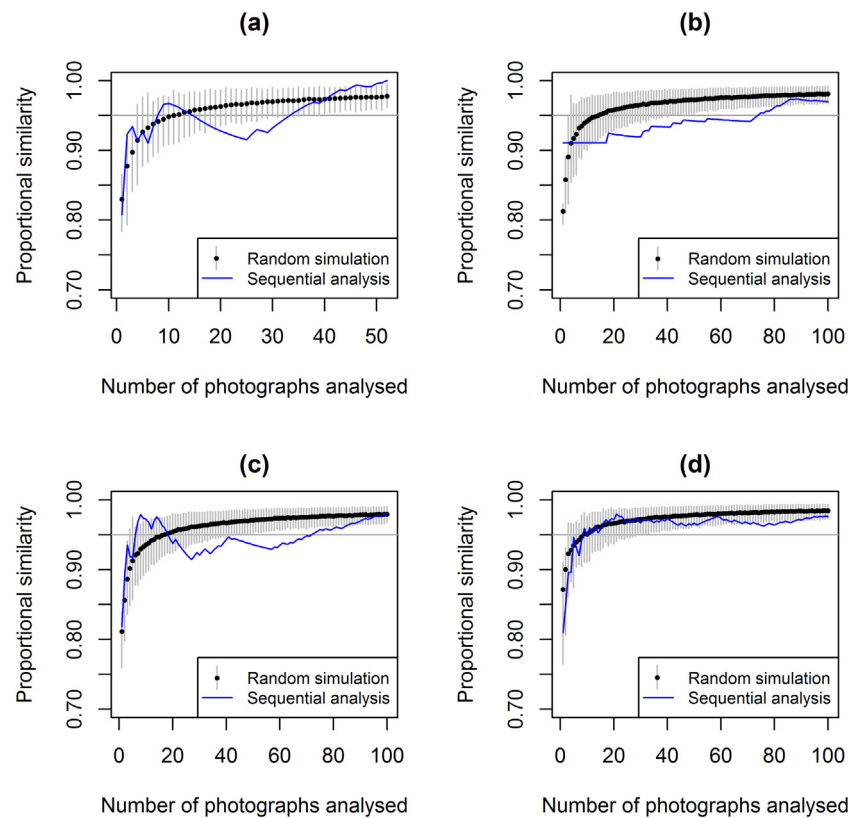


Fig. 5. Proportional similarity in photograph categorisation between simulated samples of photographs and the full dataset, for (a) Berlayer Creek, (b) Chek Jawa, (c) Mandai, and (d) SBWR. Increasing sampling effort was simulated by increasing the sample size of photographs, in increments of one from one to either the total number categorised for each site (52, in the case of Berlayer Creek) or 100 (for the other three sites). Black points indicate the mean of 1000 simulated samples. Grey vertical lines from points indicate 95% confidence intervals. The degree of 95% similarity is marked with a darker grey horizontal line. The blue line indicates the proportional similarity of a sequential analysis (classification of the first 100 photographs in the order that they were supplied by the Flickr API).

Singapore's few existing mangroves will come under increasing pressure from development in the future (Yee et al., 2010), so it is valuable to understand the diversity of CES that they provide. The contrasting sites have different cultural roles and are valued by the public for different reasons, so we cannot treat all mangrove habitat as comparable. Losing one area of mangrove as a result of development or environmental degradation may result in the loss of a resource that is not available at the remaining mangroves. To combat this risk, managers could seek to increase the diversity of cultural uses present at some of the sites, for example by replicating the diverse habitat types and infrastructure (both within and outside the mangrove) that are present at Chek Jawa.

4.2. Spatial variation in CES provision within a site

The method presented in this study was able to discriminate fine-scale variation in CES usage within a single site. Within Chek Jawa, all types of photographs were clustered on or close to the footpaths because locomotion is difficult in mangroves (Green et al., 1997), and visitors are restricted from leaving the footpaths to reduce damage. The three types of photograph showed different patterns in response to the other environmental factors; in particular to the distance from focal points and the neighbouring majority habitat type. Social recreation and landscape photographs were more likely to occur closer to the focal points, suggesting that these structures attracted people to stop and take these kinds of photographs. The effect of focal points on nature recreation photographs was less pronounced,

probably because wildlife are mobile and unpredictable, so photographs of these subjects must be taken opportunistically. It is interesting to note that the occurrence of nature recreation photographs was affected by habitat type; these photographs were much less likely in areas of open ocean than in mangrove or terrestrial areas. Boardwalks are commonly built to facilitate interactions between people and intertidal wildlife (Orams, 1996), so the lack of nature photographs around the ocean boardwalk at Chek Jawa is surprising. Environmental managers at this site may be keen to investigate whether there are few photographs taken in this area because there is no wildlife, because it is not noticeable to people, or because visitors are not interested in it. These management issues could respectively be addressed by habitat improvement, increased signage about the wildlife, or education of visitors.

4.3. The utility of rapid CES assessments using social media data

Social media data could become an important tool in CES assessment because the use of a relatively small number of freely available, spatially explicit photographs can provide a good overview of the cultural uses of a site. A 30-min investment in classifying photographs is small when compared to the time required to carry out interviews or visitor surveys such as those that are commonly employed to assess CES (Milcu et al., 2013; Hernández-Morcillo et al., 2013). There was little variation in photograph classification between people from different backgrounds, indicating that the use of straightforward, objective criteria can successfully reduce the effects of classifier bias. The framework

presented here could be integrated within ecosystem service assessment frameworks, including formal guidelines such as TESSA (Peh et al., 2013). The content analysis of social media data can add detail to existing methods that use such data as an index for recreational popularity, such as the implementation provided by InVEST (InVEST, 2014).

This study analysed objective CES categories based on photograph content, but from these usage categories it is possible to infer ecosystem service types as defined by the MA (2005). Social recreation and nature appreciation photographs represent two forms of recreation (social visits and wildlife viewing), while many of the nature and landscape photographs are likely to have been taken for aesthetic reasons. Sense of place and inspiration are less directly connected to the CES uses defined in this study, but many of the social and landscape photographs may be inspired by a feeling of sense of place; people are recording that they were here, and capturing a familiar view. Similarly, we can infer that some of the more artistic landscape photographs are a response to the inspirational quality of the panorama. Cultural heritage, existence value, and spiritual values do not require a physical interaction between an environment and visitors, so may be rarely captured by social media photographs. It is therefore important to consider these novel sources of data as a tool within a more holistic framework, rather than an all-in-one solution to assessing CES.

Analyses of social media data depend on the presence of sufficient data within the area of interest, so this approach may not be viable in remote regions, or areas with a low market penetration of cameras with geo-referencing technology. Conversely, popular areas may be difficult to assess fairly because a large number of photographs may be uploaded daily, biasing the sample returned by Flickr (the most recent 250) to the current season. We found little impact of photograph sequence on the sensitivity of the classification, but still recommend that photograph order should be randomised prior to content analysis. At popular sites it may be necessary to divide the search area into smaller patches so that multiple searches can be conducted, thus providing a larger number of photographs. Alternatively, photographs could be sampled from Flickr on multiple occasions

5. Conclusions

CES are an important, though under-quantified set of ecosystem services. Social media data could provide useful information for managing CES at multiple spatial scales, and the novel content analysis presented here may help us to understand the different CES represented in social media photographs, and thus differentiate between different cultural uses of the environment. CES indicators derived from social media could be rapid and cost-effective tools for researchers, managers and decision-makers to quickly assess CES provision at fine spatial scales, aiding the future management and conservation of culturally important ecosystems.

Acknowledgements

This research was supported by the Ministry of Education, Government of Singapore (R-109-000-147-112). We thank the seven volunteers who classified photographs, and two reviewers for their helpful feedback.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ecolind.2015.01.034>.

References

- Bennett, E.M., Peterson, G.D., Gordon, L.J., 2009. Understanding relationships among multiple ecosystem services. *Ecol. Lett.* 12, 1394–1404.
- Burkhard, B., Kroll, F., Müller, F., Windhorst, W., 2009. Landscapes' capacities to provide ecosystem services—a concept for land-cover based assessments. *Landsc. Online* 15, 1–22.
- Casalegno, S., Inger, R., Desilvey, C., Gaston, K.J., 2013. Spatial covariance between aesthetic value & other ecosystem services. *PLOS ONE* 8, e68437.
- Crossman, N.D., Burkhard, B., Nedkov, S., Willemen, L., Petz, K., Palomo, I., Drakou, E.G., Matin-Lopez, B., McPherson, T., Boyanova, K., Alkemade, R., Egoh, B., Dunbar, M.B., Maes, J., 2013. A blueprint for mapping and modelling ecosystem services. *Ecosyst. Serv.* 4, 4–14.
- Dorward, C.E., Moore, R.L., Leung, Y.-F., 2010. Visitors' perceptions of a trail environment and effects on experiences: a model for nature-based recreation experiences. *Leis. Sci.* 32, 33–54.
- EAAFP, 2012. Sungei Buloh Wetland Reserve, Singapore. East Asian-Australasian Flyway Partnership, URL <http://www.eaaflyway.net/sungei-buloh-wetland-reserve-singapore/> (accessed 03.10.14).
- Elith, J., Phillips, S.J., Hastie, T., Dudik, M., Chee, Y.E., Yates, C.J., 2011. A statistical explanation of MaxEnt for ecologists. *Divers. Distrib.* 17, 43–57.
- Fish, R.D., 2011. Environmental decision making and an ecosystems approach: some challenges from the perspective of social science. *Prog. Phys. Geogr.* 35, 671–680.
- Flickr, 2014. Flickr API Guide, URL <https://www.flickr.com/services/api/flickr.photos.search.html> (accessed October 2014).
- Friess, D.A., Phelps, J., Leong, R.C., Lee, W.K., Wee, A.K.S., Sivasothi, N., Oh, R.R.Y., Webb, E.L., 2012. Mandai mangrove, Singapore: lessons for the conservation of Southeast Asia's mangroves. *Raffles Bull. Zool.* S25, 55–65.
- Gamer, M., Lemon, J., Fellows, I., Puspendra, S., 2012. irr: Various Coefficients of Interrater Reliability and Agreement. R Package Version 0.84. <http://CRAN.R-project.org/package=irr>
- Green, E.P., Mumby, P.J., Edwards, A.J., Clark, C.D., Ellis, A.C., 1997. Estimating leaf area index of mangroves from satellite data. *Aquat. Bot.* 58, 11–19.
- Guerry, A.D., Ruckelshaus, M.H., Arkema, K.K., Bernhardt, J.R., Guannel, G., Kim, C.-K., Marsik, M., Papenfus, M., Toft, J.E., Verutes, G., Wood, S.A., Beck, M., Chan, F., Chan, K.M.A., Gelfenbaum, G., Gold, B.D., Halpern, B.S., Labiosa, W.B., Lester, S.E., Levin, P.S., McField, M., Pinsky, M.L., Plummer, M., Polasky, S., Ruggiero, P., Sutherland, D.A., Tallis, H., Day, A., Spencer, J., 2012. Modeling benefits from nature: using ecosystem services to inform coastal and marine spatial planning. *Int. J. Biodivers. Sci. Ecosyst. Serv. Manag.* 8, 107–121.
- Hayes, A.F., Krippendorff, K., 2007. Answering the call for a standard reliability measure for coding data. *Commun. Methods Meas.* 1, 77–89.
- Henderson, J.C., 2000. Managing tourism in small islands: the case of Pulau Ubin, Singapore. *J. Sustain. Tour.* 8, 250–262.
- Hernández-Morcillo, M., Plieninger, T., Bieling, C., 2013. An empirical review of cultural ecosystem service indicators. *Ecol. Indic.* 29, 434–444.
- IDA, 2014. Statistics on Telecom Services for 2012 (Jul–Dec). Infocomm Development Authority of Singapore, URL <http://www.ida.gov.sg/Infocomm-Landscape/Facts-and-Figures/Telecommunications/Statistics-on-Telecom-Services/Statistics-on-Telecom-Services-for-2012-Jul-Dec> (accessed October 2014).
- Instagram, 2014. Instagram Developer Manual, URL <http://instagram.com/developer/> (accessed October 2014).
- InVEST, 2014. InVEST Documentation – Visitation: Recreation and Tourism, URL <http://data.naturalcapitalandresilienceplatform.org/invest-releases/documentation/current-release/recreation.html>
- Landis, R., Koch, G.G., 1977. The measurement of observer agreement for categorical data. *Biometrics* 33, 159–174.
- MA, 2005. Ecosystems and Human Well-being: Biodiversity Synthesis. The Millennium Ecosystem Assessment. Island Press, Washington, DC.
- Milcu, A.I., Hanspach, J., Abson, D., Fischer, J., 2013. Cultural ecosystem services: a literature review and prospects. *Ecol. Soc.* 18, 44.
- Nahuelhual, L., Carmona, A., Lozada, P., Jaramillo, A., 2013. Mapping recreation and ecotourism as a cultural ecosystem services: an application at the local level in Southern Chile. *Appl. Geogr.* 40, 71–82.
- Norton, L.R., Inwood, H., Crowe, A., Baker, A., 2012. Trialling a method to quantify the cultural services of the English landscape using Countryside Survey data. *Land Use Policy* 29, 449–455.
- Orams, M.B., 1996. A conceptual model of tourist–wildlife interaction: the case for education as a management strategy. *Aust. Geogr.* 27, 39–51.
- Peh, K.S.-H., Balmford, A., Bradbury, R.B., Brown, C., Butchart, S.H.M., Hughes, F.M.R., Stattersfield, A., Thomas, D.H.L., Walpole, M., Bayliss, J., Gowing, D., Jones, J.P.G., Lewis, S.L., Mulligan, M., Pandeya, B., Stratford, C., Thompson, J.R., Turner, K., Vira, B., Willcock, S., Birch, J.C., 2013. TESSA: a toolkit for rapid assessment of ecosystem services at sites of biodiversity conservation importance. *Ecosyst. Serv.* 5, e51–e57.
- Phillips, S.J., Dudik, M., 2008. Modeling of species distributions with MaxEnt: new extensions and a comprehensive evaluation. *Ecography* 31, 161–175.
- Pleasant, M.M., Gray, S.A., Lepczyk, C., Fernandes, A., Hunter, N., Ford, D., 2014. Managing cultural ecosystem services. *Ecosyst. Serv.* 8, 141–147.
- Plieninger, T., Dijk, S., Oteros-Rozas, E., Bieling, C., 2013. Assessing, mapping, and quantifying cultural ecosystem services at community level. *Land Use Policy* 33, 118–129.
- Sherrouse, B.C., Semmens, D.J., 2012. Social Values for Ecosystem Services, Version 2.0: Documentation and User Manual. U.S. Geological Survey Open-File Report 2012-1023.

- Villa, F., Ceroni, M., Bagstad, K., Johnson, G., Krivov, S., 2009. [ARIES: a new tool for ecosystem services assessment, planning, and valuation](#). In: 11th Annual BIOECON Conference on Economic Instruments to Enhance the Conservation and Sustainable Use of Biodiversity, Venice, Italy.
- Wood, S.A., Guerry, A.D., Silver, J.M., Lacayo, M., 2013. [Using social media to quantify nature-based tourism and recreation](#). *Sci. Rep.* 3, 2976.
- Yee, A.T.K., Ang, W.F., Teo, S., Liew, S.C., Tan, H.T.W., 2010. [The present extent of mangrove forests in Singapore](#). *Nat. Singap.* 3, 139–145.
- Zandbergen, P.A., Barbeau, S.J., 2011. [Positional accuracy of assisted GPS data from high-sensitivity GPS-enabled mobile phones](#). *J. Navig.* 64, 381–399.